Chhavi Nayyar

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EDUCATION

University of British Columbia, Vancouver, BC Sept 2022 – May 2027

BSc, Combined Major in Computer Science, Biology, & Cognitive Systems

Leadership: Design Director, UBC UX Hub · Campus Ambassador, Adobe · Z Student Ambassador, IBM · Member, Women in CS

EXPERIENCE

Mastercard, Vancouver, BC May 2026 – Aug 2026

Software Engineer Intern

- Architected and shipped backend microservices (Java, Python; gRPC/REST) for low-latency, high-throughput transaction processing within a large-scale production payments platform, writing modular, testable code within existing team frameworks and abstractions
- Deployed and evaluated ML fraud-detection pipelines (Docker, Terraform, AWS) in production; raised code quality by adding test coverage and observability instrumentation, enabling faster and safer iteration across the codebase
- Built internal dashboards surfacing fraud-detection and transaction data for engineering teams, and contributed to client-facing tools used by bank partners and their end customers

BC Cancer (Nunez Lab), Vancouver, BC June 2025 – Present

Software Developer

- Shipped a patient-partner portal over a 951-page production resource database (Cancer Navigator): full-text + smart search, 6 filterable columns with removable chips, and URL-dedup logic preventing duplicate entries; routed partner-submitted edit/broken-link reports into a moderation queue for the FLEX student team; extended the admin app with a new activity tab and live-API-backed metadata fields
- Engineered unsupervised ML pipelines (Python/scikit-learn) to analyze qualitative healthcare feedback, improving thematic pattern detection by 40%; built and shipped an open-source concept-mapping toolkit (NetworkX, FastAPI, Plotly) cutting manual analysis time by 50%; co-first-authored resulting publication
- Built pipeline components for an LLM-based depression/bipolar support chatbot (PolarUS/DEPBOT), including direct patient-facing interaction testing; reduced AWS cloud spend by $260/month by identifying and eliminating unused services

RESEARCH EXPERIENCE

UBC Department of Computer Science, Vancouver, BC Apr 2025 – Apr 2026

Deep Learning Engineer + Teaching Assistant – Human-Centered AI & Cognitive Systems

- Architected and evaluated VTNet models (TensorFlow, PyTorch) for real-time eye-tracking analysis, achieving a 25% improvement in predictive accuracy, with distributed training on Slurm HPC clusters accelerating pipelines by 30%
- Supported graduate instruction and led lab sessions for CPSC 532C/554C and COGS 300/303, facilitating seminars on algorithmic fairness and bridging cognitive science theory with applied AI implementation

PUBLICATIONS

Nayyar, C.*, Xu, H.*, Hilbers, D., Avery, J., Raman, S., Bates, A.T., Conati, C., Fayaz-Bakhsh, A., & Nunez, J. (2026). Towards clinical implementation of artificial intelligence in cancer care: Concept mapping analysis of provincial workshop findings. Implementation Science Communications. doi:10.1186/s43058-026-01075-x · *co-first author

PROJECTS

Your Search Box: Multi-Modal AI Search Service | TypeScript, Next.js, Node.js, NLP

- Built a server-side rendered, real-time search platform (TypeScript, Next.js) supporting voice, image, and text input, with an adapter pattern for headless CMS integration (Shopify, WordPress); live commercial product in production

WebAbility.io: AI-Powered Accessibility Platform | TypeScript, Node.js, React, Playwright, axe-core

- Built a live accessibility-as-a-service product: a scanning engine (Playwright + axe-core + 60+ custom WCAG 2.1/2.2 detectors) that identifies violations and uses an LLM to generate automated fixes

SKILLS & INTERESTS

- Languages: Python, Java, TypeScript, JavaScript, C++, SQL, HTML/CSS
- AI/ML: TensorFlow, PyTorch, scikit-learn, LLM pipelines & prompt engineering, NetworkX, Plotly, ML pipeline evaluation
- Backend & APIs: Node.js, Express.js, Flask, FastAPI, gRPC, REST, Microservices
- Cloud & DevOps: AWS (EC2, S3, cost optimization), Docker, Terraform, CI/CD, Linux, Slurm
- Frontend & Data: React, Next.js, Angular, Tailwind CSS, PostgreSQL, MySQL, MongoDB, NoSQL